

Your task is to assign CS classes to Professors. There are 5 classes and 3 professors.

You are constrained by the fact that each professor can only teach one class at a time.

The classes and their fixed times are:

Class 1 - Intro to Programming: meets from 8:00-9:00am

Class 2 - Intro to Artificial Intelligence: meets from 8:30-9:30am

Class 3 - Natural Language Processing: meets from 9:00-10:00am

Class 4 - Computer Vision: meets from 9:00-10:00am

Class 5 - Machine Learning: meets from 10:30-11:30am

The professors are:

Professor A, who is qualified to teach Classes 1, 2, and 5.

Professor B, who is qualified to teach Classes 3, 4, and 5.

Professor C, who is qualified to teach Classes 1, 3, and 4.

a) Formulate this problem as a CSP in terms of **variables, **domains**, and **constraints**.**

Express your answers using mathematical, set, and logical.

Variables Domains (or unary constraints)

C_1 $\{A, C\}$

C_2 $\{A\}$

C_3 $\{B, C\}$

C_4 $\{B, C\}$

C_5 $\{A, B\}$

Binary Constraints

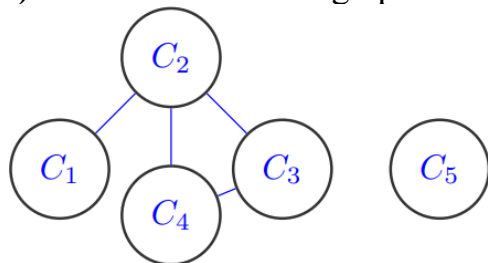
$C_1 \neq C_2$

$C_2 \neq C_3$

$C_2 \neq C_4$

$C_3 \neq C_4$

b) Draw the constraint graph.



c) Using the Degree heuristic, which variable would be assigned first?

You want to invite five guests: (A)hmed, (B)andar, (C)hahed, (D)hafer, and (E)rfan. You want to setup the menu and seating arrangements.

You have three tables labelled as {1, 2, 3}.

You have prepared the following menu options: {chicken, camel, lamb, fish}.

For each guest, you want to assign a table and menu option.

You have the following constraints:

- Ahmed only likes fish.
- Bandar only likes chicken and camel.
- Ahmed and Chahed argue with each other, so do not seat them at same table.
- Bandar and Dhafer are friends, so seat them at same table. Also, they will share their food, so they should get different menu options.
- Erfan wants to sit alone.

Hint: A variable X can be assigned a tuple (table, menu). Use notation X[0] and X[1] to refer to table and menu assignments, respectively.

a) Formulate this problem as a CSP in terms of **variables, domains, and constraints**. Express your answers using mathematical, set, and logical.

Variables are: A, B, C, D, E representing the five guests.

Domains: all variables is a 2-tuple (table, menu) where $table \in \{1, 2, 3, 4\}$ and $menu \in \{chicken, camel, lamb, fish\}$.

Constraints:

$$A[1] = fish$$

$$(B[1] = chicken) \text{ or } (B[1] = camel)$$

$$A[0] \neq C[0]$$

$$B[0] = D[0]$$

$$B[1] \neq D[1]$$

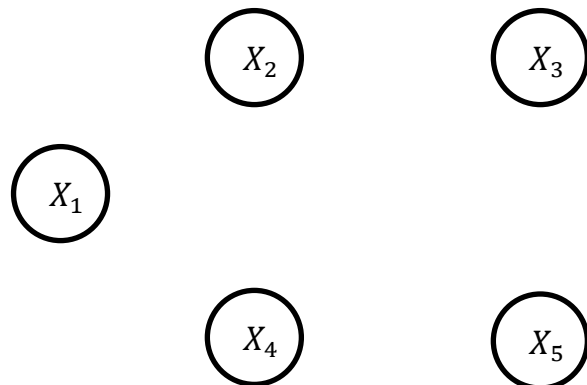
For the last constraint, it is a straightforward alldiff for E:

$$(E[0] \neq A[0]) \text{ and } (E[0] \neq B[0]) \text{ and } (E[0] \neq C[0]) \text{ and } (E[0] \neq D[0])$$

b) Draw the constraint graph.

Consider the following CSP problem with variables X_1, X_2, X_3, X_4, X_5 each with domain $D = \{1, 2, 3, 4, 5, 6\}$. The following are the constraints:

$X_1 < 5$
 $X_5 \neq 6$
 $X_1 \neq X_5$
 $X_1 \geq X_2$
 $X_2 > X_3$
 $X_3 < X_5$



(a) Draw the constraint graph for this CSP.

(b) Suppose you set $X_5 = 2$. In the table below, cross out values to **enforce unary constraints, then run forward-checking**.

Give your final answer in table on the right by filling in the corresponding circles of **crossed out** values (values to **remove** from the domain).

X_1	1	2	3	4	5	6
X_2	1	2	3	4	5	6
X_3	1	2	3	4	5	6
X_4	1	2	3	4	5	6
X_5		2				

X_1	☐	☐	☐	☐	☐	☐
X_2	☐	☐	☐	☐	☐	☐
X_3	☐	☐	☐	☐	☐	☐
X_4	☐	☐	☐	☐	☐	☐
X_5		☐				

(c) Suppose we took the situation from the answer in (b), and we set $X_3 = 1, X_5 = 2$. Now which unassigned variable would MRV prefer? If there are ties, write them all.

(d) In the table below, **enforce unary constraints, THEN enforce arc consistency** on the **entire CSP**.

X_1	1	2	3	4	5	6
X_2	1	2	3	4	5	6
X_3	1	2	3	4	5	6
X_4	1	2	3	4	5	6
X_5	1	2	3	4	5	6

X_1	○	○	○	○	○	○
X_2	○	○	○	○	○	○
X_3	○	○	○	○	○	○
X_4	○	○	○	○	○	○
X_5	○	○	○	○	○	○

(e) Suppose you set $X_2 = 4$. In the table below, **cross out values to enforce unary constraints, THEN arc consistency on the entire CSP**.

X_1	1	2	3	4	5	6
X_2				4		
X_3	1	2	3	4	5	6
X_4	1	2	3	4	5	6
X_5	1	2	3	4	5	6

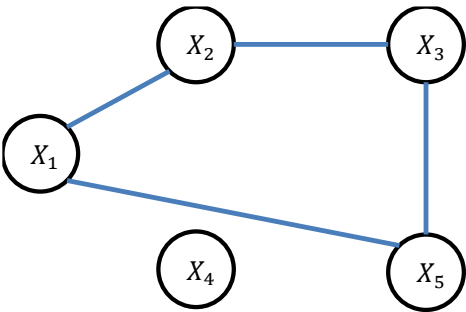
X_1	○	○	○	○	○	○
X_2						
X_3	○	○	○	○	○	○
X_4	○	○	○	○	○	○
X_5	○	○	○	○	○	○

(f) Suppose we took the situation from the answer in (e), and we set $X_1 = 4, X_2 = 4$. Now let's say we selected X_5 for variable assignment, using LCV what is the value ordering for X_5 ?

Solution of previous question

Consider the following CSP problem with variables X_1, X_2, X_3, X_4, X_5 each with domain $D = \{1, 2, 3, 4, 5, 6\}$. The following are the constraints:

- $X_1 < 5$
- $X_5 \neq 6$
- $X_1 \neq X_5$
- $X_1 \geq X_2$
- $X_2 > X_3$
- $X_3 < X_5$



- (a) Draw the constraint graph for this CSP.
- (b) Suppose you set $X_5 = 2$. In the table below, cross out values to **enforce unary constraints, then run forward-checking**. Give your final answer in table on the right by filling in the corresponding circles of **crossed out** values (values to **remove** from the domain).

X_1	1	2	3	4	5	6
X_2	1	2	3	4	5	6
X_3	1	2	3	4	5	6
X_4	1	2	3	4	5	6
X_5		2				

X_1	☐	☒	☐	☐	☒	☒
X_2	☐	☐	☐	☐	☐	☐
X_3	☐	☒	☒	☒	☒	☒
X_4	☐	☐	☐	☐	☐	☐
X_5		☐				

- (c) Suppose we took the situation from the answer in (b), and we set $X_3 = 1, X_5 = 2$. Now which unassigned variable would MRV prefer? If there are ties, write them all.

(d) In the table below, **enforce unary constraints, THEN enforce arc consistency** on the **entire CSP**.

X_1	1	2	3	4	5	6
X_2	1	2	3	4	5	6
X_3	1	2	3	4	5	6
X_4	1	2	3	4	5	6
X_5	1	2	3	4	5	6

X_1	☐	☐	☐	☐	☐	☐
X_2	☐	☐	☐	☐	☐	☐
X_3	☐	☐	☐	☐	☐	☐
X_4	☐	☐	☐	☐	☐	☐
X_5	☐	☐	☐	☐	☐	☐

(e) Suppose you set $X_2 = 4$. In the table below, **cross out values to enforce unary constraints, THEN arc consistency** on the **entire CSP**.

X_1	1	2	3	4	5	6
X_2				4		
X_3	1	2	3	4	5	6
X_4	1	2	3	4	5	6
X_5	1	2	3	4	5	6

X_1	☐	☐	☐	☐	☐	☐
X_2						
X_3	☐	☐	☐	☐	☐	☐
X_4	☐	☐	☐	☐	☐	☐
X_5	☐	☐	☐	☐	☐	☐

(f) Suppose we took the situation from the answer in (e), and we set $X_1 = 4, X_2 = 4$. Now let's say we selected X_5 for variable assignment, using LCV what is the value ordering for X_5 ?

You should give a list of values, ordered from least-constraining to most-constraining. Note that you just check **unassigned** neighbors; and no need to propagate (we are not doing ac3).

$X_5 = 2$ will remove 2 values from X_3 domain

$X_5 = 3$ will remove 1 value from X_3 domain

$X_5 = 5$ will remove 0 values from X_3 domain

So value ordering for X_5 is [5, 3, 2]

Consider a CSP with 5 variables X_1, X_2, X_3, X_4, X_5 each with domain $\{0, 1, 2, 3, 4, 5\}$. Here are the constraints:

$$X_4 < 1$$

$$X_5 \leq 1$$

$$X_2 > X_3$$

$$X_3 < X_5$$

$$X_4 < X_5$$

a) In the table below, **enforce unary constraints, THEN enforce arc consistency** on the entire CSP. Use table on left to work out the answer.

Give your final answer in table on the right by filling in the corresponding circles of **crossed out** values (values to remove from the domain).

x_1	0	1	2	3	4	5
x_2	0	1	2	3	4	5
x_3	0	1	2	3	4	5
x_4	0	1	2	3	4	5
x_5	0	1	2	3	4	5

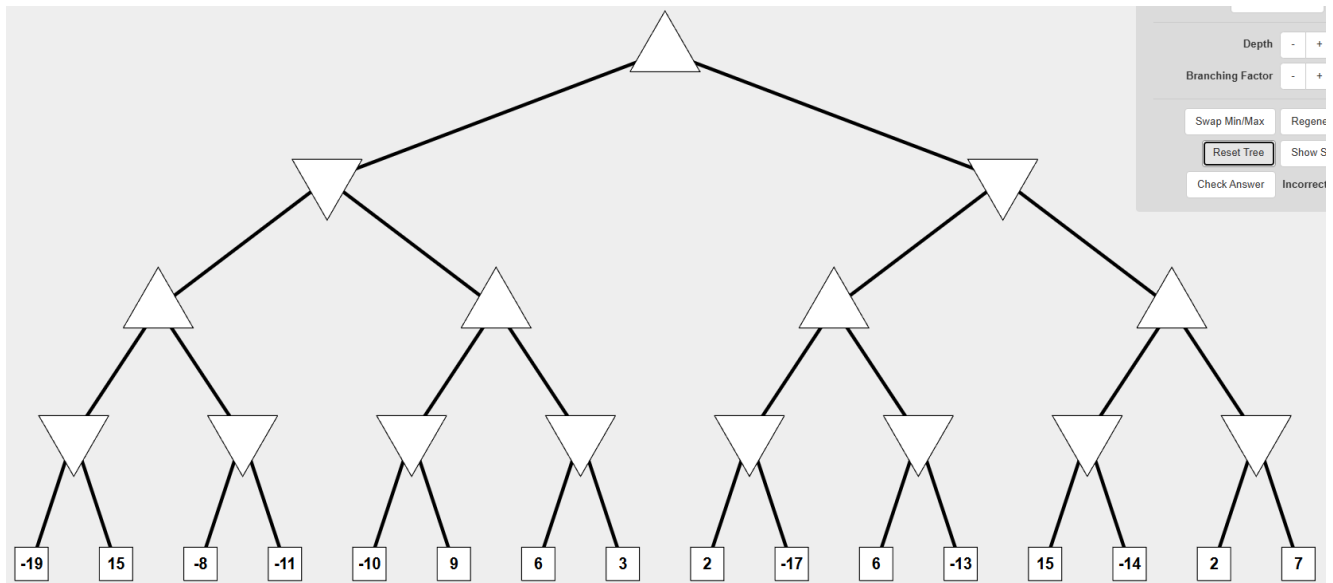
x_1	☐	☐	☐	☐	☐	☐
x_2	☐	☐	☐	☐	☐	☐
x_3	☐	☐	☐	☐	☐	☐
x_4	☐	☐	☐	☐	☐	☐
x_5	☐	☐	☐	☐	☐	☐

b) Using your answer for (a), according to the Minimum Remaining Value (MRV) heuristic, which variable should be assigned first? If there are ties, pick the first-ordered one (order as given in question).

X_3

c) Using your answer for (a), according to the Degree heuristic, which variable should be assigned first? If there are ties, pick the first-ordered one (order as given in question).

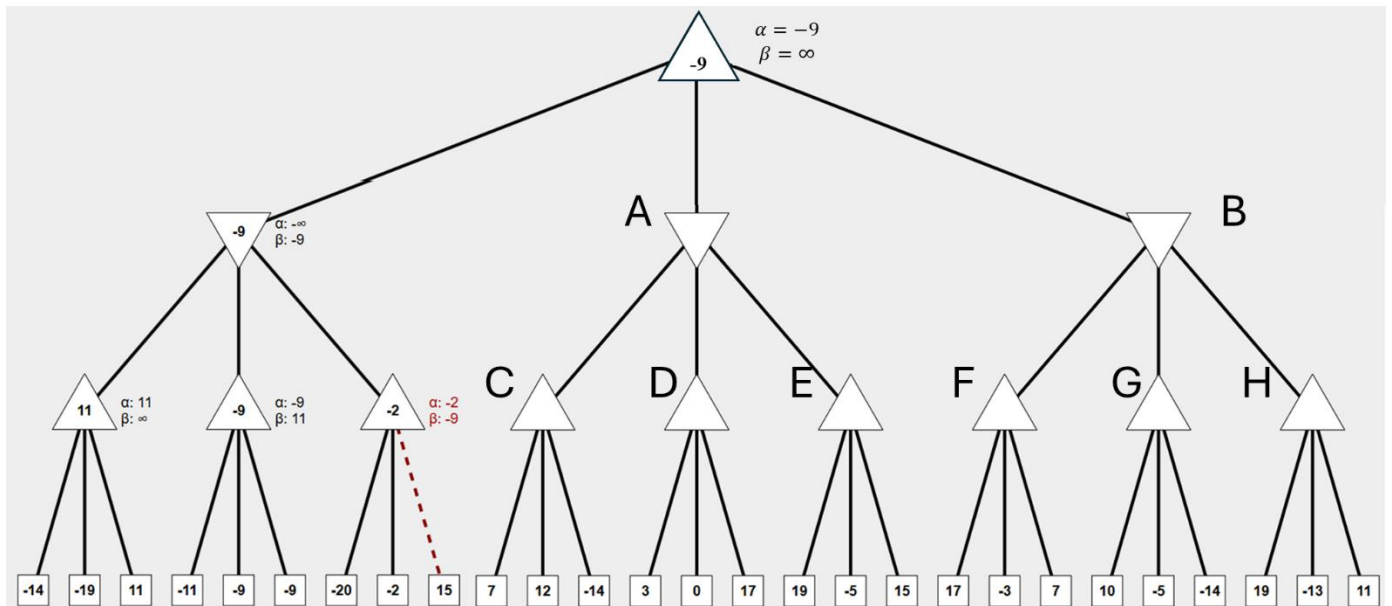
X_3



Utility at terminal nodes are for the MAX player. For the game-tree above:

- Apply minimax. Provide the **value** of each game node. After you are done, which action should be taken by MAX? (left or right branch)
- Apply minimax with alpha-beta pruning. After you are done, at each node provide the **value, alpha, and beta**. Process nodes in **left-to-right order**.

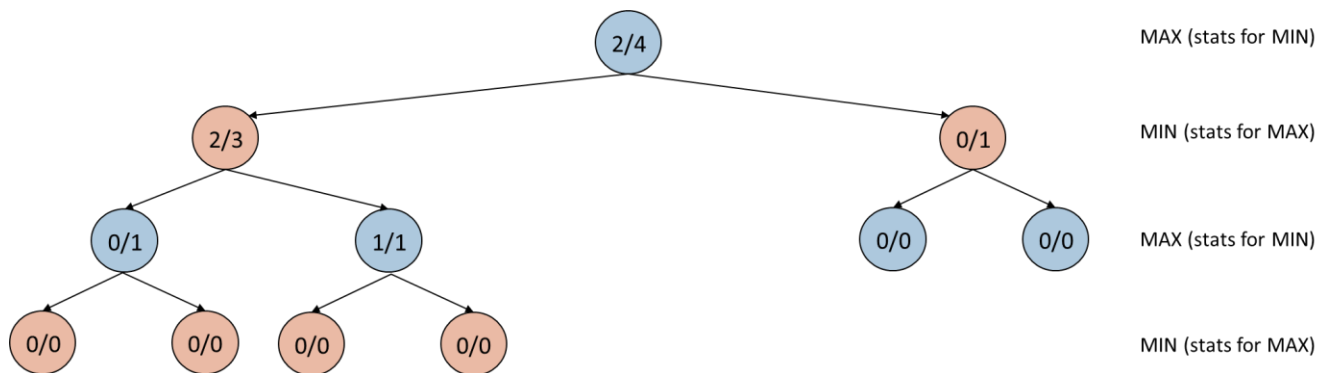
To see the solution, and to generate your own game-trees for practice, see <https://pasccha.ch/info2/abTreePractice/>



For the game-tree above, alpha-beta minimax has been applied and the left subtree from the root node has already been processed. Continue running alpha-beta minimax until completion. Process nodes in **left-to-right** order. **Show** your work on the tree. **After** you are done, fill your answers in the table below. For nodes that are completely pruned and have no values, then just write **pruned**.

Node	<i>value</i>	α	β	Which child is pruned? (left, middle, right, all, none)
A	12	-9	12	none
B	10	12	10	right
C	12	12	∞	none
D	17	17	12	none
E	19	19	12	middle and right
F	17	17	∞	none
G	10	12	17	none
H	pruned	pruned	pruned	pruned

In the real-world it is MAX's turn. Suppose MAX wants to apply MCTS. The following is the current state of the game-tree. The root node shows the win stats for MIN, then every level in the tree alternates between win stats for MAX and MIN.



a) Suppose that MCTS is run for **three more** iterations, and the simulation results are as follows: MAX win, MAX win, MIN win.

Grow the tree to reflect what MCTS would do and update the stats. Assume that the game has only two actions from any state.

b) Continuing from your answer in the (a), what action will MAX select (left or right branch)?

Recall the heuristic minimax algorithm. Suppose we have a game that ends in d turns, and we want to cutoff the search at depth l .

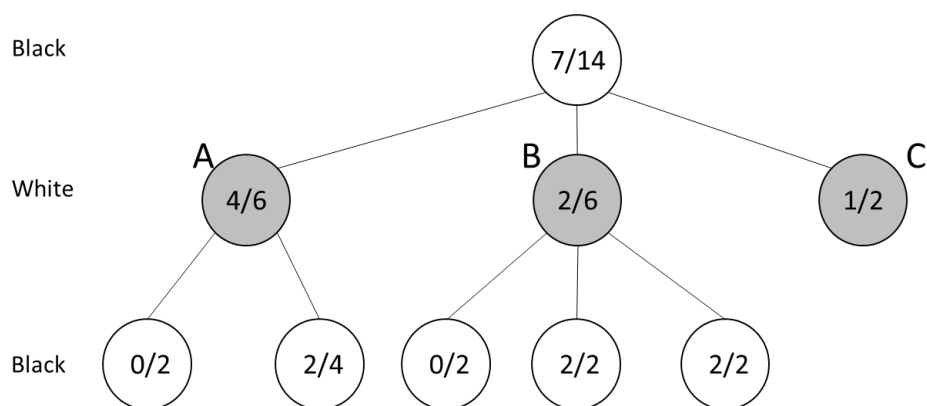
Suppose you use an imperfect/error heuristic. Is h-minimax guaranteed to be optimal?

How would you design a *perfect* heuristic? That is, a heuristic that returns the exact win probability of a state.

Recall the UCB1 formula.

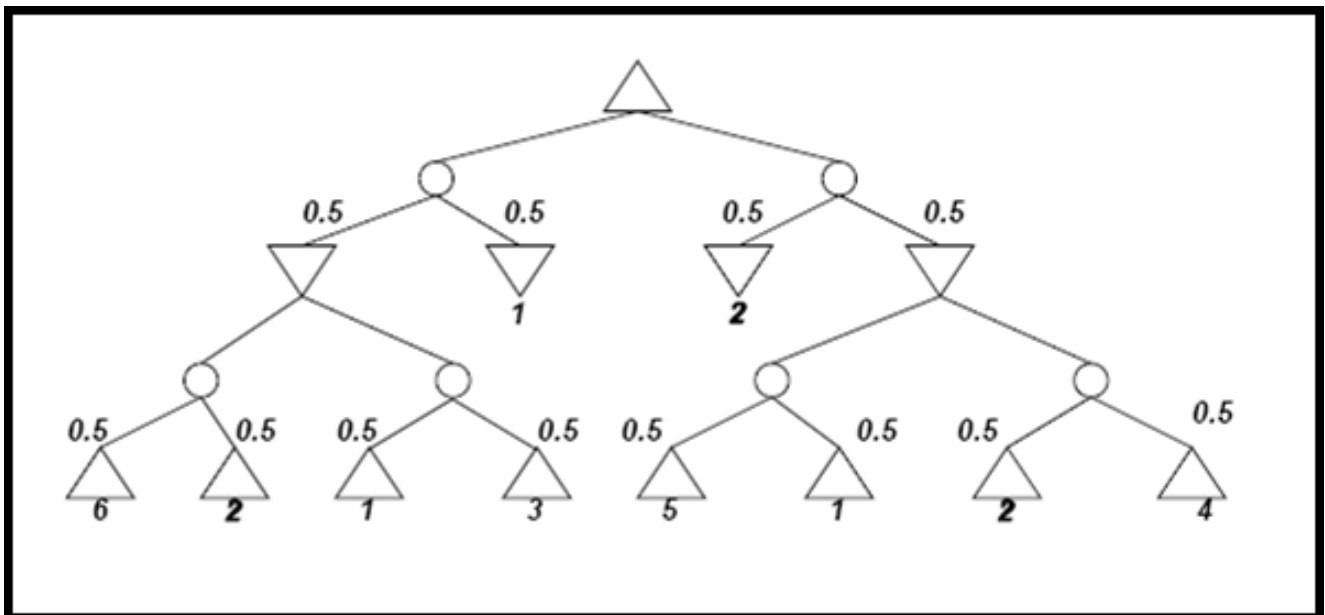
$$UCB1(n) = \frac{U(n)}{N(n)} + C \times \sqrt{\frac{\ln N(\text{PARENT}(n))}{N(n)}}$$

In the real-world it is Black's turn. Suppose Black wants to apply MCTS. The root node shows the win stats for White, then every level in the tree alternates between win stats for Black and White. Only 3 possible actions from any state. When tie-breaking, select left-most child first.



a) Which leaf node will be selected for running a simulation? Justify your answer with computation and clear explanations.

Consider the following game with a binary event.



a) Perform expectiminimax. Provide the **value** of each node.

b) What action should MAX take? (Left or right branch)

Consider the following KB:

$$R_1: A \wedge B \Rightarrow C$$

Does $KB \models [(A \Rightarrow B) \vee (B \Rightarrow C)]$?

Using model-checking to check if entailment holds.

A	B	C					
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Consider the following KB:

$$R_1: R \vee T$$

$$R_2: Q \wedge U$$

$$R_3: (S \vee T) \Rightarrow (Q \Rightarrow P)$$

$$R_4: \neg R \vee S$$

Does $KB \models P$?

Clearly show a direct-proof. For each of your steps, write the name of the rule you used on the right and the sentences involved in the rule. For example, “Resolution with R_9 and R_{14} ”. For each new clause you generate, give it a label R_x

$$R_5: (S \vee T)$$

Resolution with R_1 and R_4

$$R_6: Q \Rightarrow P$$

Modus ponens R_3 and R_5

$$R_7: Q$$

From R_2 using and-simplification

$$R_8: P$$

Modus ponens R_6 and R_7

Consider the following KB:

$$R_1: A \vee B$$

$$R_2: \neg C \vee \neg D \vee E$$

$$R_3: C$$

Does $KB \models [(A \vee B) \wedge (\neg D \vee E)]$?

Clearly show a direct-proof. For each of your steps, write the name of the rule you used on the right and the sentences involved in the rule. For example, “Resolution with R_4 and R_5 ”. For each new clause you generate, give it a label R_x

Consider the following KB:

$$R_1: (P \wedge Q) \Rightarrow (R \vee S)$$

$$R_2: (R \wedge Q) \Rightarrow (\neg P \vee S)$$

$$R_3: \neg S \Rightarrow Q$$

Does $KB \models (P \Rightarrow S)$?

Clearly show a direct-proof. For each of your steps, write the name of the rule you used on the right and the sentences involved in the rule. For example, “Resolution with R_4 and R_5 ”. For each new clause you generate, give it a label R_x

Consider the following CNF KB:

$$S1: (\neg A \vee B \vee E)$$

$$S2: (\neg E \vee D).$$

$$S3: (\neg C \vee \neg F \vee \neg B)$$

$$S4: (\neg E \vee B).$$

$$S5: (\neg B \vee F).$$

$$S6: (\neg B \vee C).$$

$$S7: (\neg B \vee A)$$

$$S8: (\neg E \vee A)$$

Use resolution to show that $KB \models (\neg A \wedge \neg B)$.

Show in clear steps that the empty clause is generated using the resolution-based proof. For each new clause you generate, give it a label SX .

Consider the KB shown below.

$$A1: A \wedge B \implies D$$

$$A2: A \wedge F \implies G$$

$$A3: B \wedge C \implies F$$

$$A4: D \wedge C \implies H$$

$$A5: A \wedge C \implies E$$

$$A6: G \wedge E \wedge H \implies I$$

$$A7: A$$

$$A8: B$$

$$A9: C$$

a) Draw the corresponding AND-OR graph for this KB.

b) Show that $KB \models I$ using forward chaining. Give your answer in steps of what is being proved. Maintain a set of actives = {} as you progress in each step.

Show your answer in steps.

Initially actives are the facts. Actives = {A, B, C}. Steps are as follows:

- A and B active, this proves D (A1). Actives = {A, B, C, D}
- A and C active, this proves E (A5). Actives = {A, B, C, D, E}
- D and C active, this proves H (A4). Actives = {A, B, C, D, E, H}
- B and C active, this proves F (A3). Actives = {A, B, C, D, E, F, H}
- A and F active, this proves G (A2). Actives = {A, B, C, D, E, F, G, H}
- G, E, and H active, this proves I (A6). Actives = {A, B, C, D, E, F, G, H, I}

I is activated, thus $KB \models I$

c) Show that $KB \models I$ using backward chaining. Give your answer in steps of what is being proved. Maintain sets already-proved = {} and sub-goals = {} as you progress.

Show your answer in steps.

Initially already-proved = {A, B, C} and sub-goals = {I}. Steps are as follows:

- I is in A6. already-proved = {A, B, C} and sub-goals = {I, G, E, H}.
- E is in A5, with A and C proved, which proves E. already-proved = {A, B, C, E} and sub-goals = {I, G, H}.
- G is in A2, A already proved, but F needs proof. already-proved = {A, B, C, E} and sub-goals = {I, G, H, F}.
- F in A3, with B and C proved, which proves F. already-proved = {A, B, C, E, F} and sub-goals = {I, G, H}.
- G in A2, and A and F proved, which proves G. already-proved = {A, B, C, E, F, G} and sub-goals = {I, H}.
- H in A4, and D requires proof. already-proved = {A, B, C, E, F, G} and sub-goals = {I, D, H}.
- D in A1, and A and B proved, which proves D. already-proved = {A, B, C, D, E, F, G} and sub-goals = {I, H}.
- H now proved. already-proved = {A, B, C, D, E, F, G, H} and sub-goals = {I}.
- I now proved. already-proved = {A, B, C, D, E, F, G, H, I} and sub-goals = {}.

Thus $KB \models I$

Consider the CNF formula:

$$(p \vee q \vee r \vee s) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg q \vee \neg r \vee s) \wedge (p \vee \neg q \vee r \vee s) \\ \wedge (q \vee \neg r \vee \neg s) \wedge (\neg p \vee \neg r \vee s) \wedge (\neg p \vee \neg s) \wedge (p \vee \neg q)$$

a) Apply $s = \text{True}$, and simplify the formula?

$$(\neg p \vee q \vee \neg r) \wedge (q \vee \neg r) \wedge (\neg p) \wedge (p \vee \neg q)$$

b) Continuing from part a), keep applying unit-propagation.

We have one unit clause above, $(\neg p)$, so we will force $p = \text{false}$, simplify, and propagate.

$$(q \vee \neg r) \wedge (\neg q)$$

Now we have $(\neg q)$ unit clause, so we force $q = \text{false}$:

$$(\neg r)$$

Now we have $(\neg r)$ unit clause, so we force $r = \text{false}$, simplifies the formula to:

$$\text{True}$$

So with this, we have found a satisfying assignment for the given formula, namely:

$$\{s = \text{true}, p = \text{false}, q = \text{false}, r = \text{false}\}$$

c) Continuing from part a), keep applying pure-literal-assignment.

From part a), we only have on pure literal $\neg r$, so we force $r = \text{false}$.

$$(\neg p) \wedge (p \vee \neg q)$$

Now we have pure literal $\neg q$, so we force $q = \text{false}$.

$$(\neg p)$$

Now we have pure literal $\neg p$, so we force $p = \text{false}$.

$$\text{True}$$

So with this, we have found a satisfying assignment for the given formula, namely:

$$\{s = \text{true}, p = \text{false}, q = \text{false}, r = \text{false}\}$$